

EDUCATION

Northeastern University <i>Master of Science, Computer Science GPA:(4.0/4.0)</i>	San Jose, United States <i>Jan. 2024 – May. 2026</i>
National Taiwan Normal University <i>Master of Science, Physics</i>	Taipei, Taiwan <i>Sep. 2020 – July. 2022</i>

SKILLS

Programming Languages: Python, Golang(Go), C, Java, JavaScript, SQL
Frameworks & Libraries: Django, Flask, FastAPI, Gin, React, TensorFlow
Databases & Caching: MySQL, Redis, DynamoDB
Cloud & DevOps: Docker, Kubernetes, GCP (Cloud Run, Compute Engine, Cloud Storage), AWS (EC2, S3, ECS), Terraform, Git, Linux, CI/CD
Certifications: Certified Kubernetes Application Developer (CKAD)

EXPERIENCE

Graduate Researcher <i>Institute of Physics, Academia Sinica</i>	Jul. 2020 – Jul. 2022 <i>Taipei, Taiwan</i>
- Developed image segmentation pipeline for single-cell tracking using Python and TensorFlow, integrating classical image processing with deep learning models (Mask R-CNN, U-Net). - Benchmarked three models on stratified datasets, achieving up to 23% improvement in F1-score with consistent gains across difficulty levels; optimized U-Net inference to 0.007s/image (69× speedup) over Mask R-CNN. - Built automated validation pipeline processing 1,600+ cell images, implementing quality checks that reduced manual review overhead by 60%	

PROJECTS

KubeRay Open Source (2.3K+ stars) <i>Go, Kubernetes, Client-Go, Cobra, Ginkgo</i>	Jul. 2025 – Present
- Implemented autoscaling replica controls for kubectrl-ray plugin, adding min/max boundary enforcement and validation logic to prevent unsafe scaling operations across distributed RayClusters. - Implemented <code>/tasks/timeline</code> API endpoint for History Server enabling task execution timeline queries from MinIO-stored historical data in Chrome Tracing Format to support post-cluster-deletion profiling workflows. - Overhauled RayCronJob testing infrastructure by adding Kubebuilder markers, regenerating CRDs and typed client code, implementing Ginkgo-based e2e tests, and integrating into Buildkite CI pipeline.	
Distributed Rate Limiter <i>Go, Redis, Lua, Docker</i>	Jul. 2025 – Sep. 2025
- Designed a high-performance distributed rate-limiting service handling 8300+ req/s with <30ms p99 latency using atomic Lua scripts ensuring consistency across multi-bucket (IP/user/global) rules. - Implemented horizontally scalable architecture using shared Redis state and stateless Go nodes, supporting safe hot reload of configuration. - Built integration tests with testcontainers-go and benchmark suite; achieved 80%+ coverage.	
Invoice Processing Automation System <i>FastAPI, JavaScript, n8n, OpenAI API, OCR</i>	May. 2025 – Jun. 2025
- Built end-to-end invoice extraction system supporting 100+ formats (PDF/JPG/PNG), reducing manual workload by 90%. - Developed fuzzy matching algorithm achieving high accuracy in product identification, handling name variations and multi-language text processing. - Designed RESTful API with FastAPI and integrated n8n workflow automation, implementing comprehensive error handling and logging mechanisms.	
SQL Learning Platform <i>Django, React, MySQL, JWT, GCP</i>	Jan. 2025 – Apr. 2025
- Built full-stack SQL learning platform with Django REST Framework (JWT auth), React frontend, and GCP deployment, supporting 300+ user submissions across 20+ SQL challenges. - Designed secure SQL sandbox using ephemeral MySQL schemas per session, isolating user queries with <2s execution latency; integrated GPT-4 API for automated solution generation and personalized error analysis. - Implemented instructor dashboard with AI-driven insights; enabled real-time updates by migrating problem data to Google Cloud Storage, reducing deploy time from 30 minutes to instant.	